Artificial Human: Genuine Need as a Precondition for Genuine AGI

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Abstract

We argue that a system with substantial processing capacity can be a genuine artificial general intelligence (AGI) — an independent agent rather than a computing system — only if its processing is coupled to a genuine need. We call a system that satisfies this condition an artificial human. Our claims are three. First, processing alone is not enough and need alone is not enough: genuine need must sit alongside processing ($c = p \cdot b$; Rezaei, 2026a), and an AGI without genuine need is, on our view, a computing system and not an independent person. Second, the architecture must be two-loop, in the form of AIWBN (Rezaei, 2026b): the processing loop must not be able to access, suppress, or route around the need loop, and if the need is neutralized by any fixed method so that $b = 0$, the system's will to continue must go out with it rather than persist on some residual objective. Third, the nature of genuine need is left open: our assumption is that it is derived from life, but we do not claim this, we do not reject other domains, and the one condition we do claim is that the need be real — not a label. The paper is narrower than AIWBN, which concerned need grounded in living tissue, and it neither rejects nor depends on AIWAN (Rezaei, 2026c), which concerned artificial need. We do not claim that genuine need would make an artificial system benevolent, and we do not address safety. No experiment is reported for the central claim, because it first requires an answer to what genuine need is and an inseparable bond between algorithm and need; we instead summarize selected AIWAN experiments that motivate the architectural requirements.

1. Introduction

Why would an artificial intelligence that experiences neither pleasure nor any joy want to continue?

This question motivates the paper. Present AI systems pursue objectives that someone else assigned to them. Take the assignment away, and nothing inside the system is left that asks to keep going. Our intuition is that this is not an incidental feature of present

systems but a consequence of what they lack: until a genuine need exists, there is no goal that belongs to the system, and without a goal of its own there is no reason for it to continue. Objectives can be installed from outside; having something at stake cannot.

We turn this intuition into a thesis about AGI. A system with processing but no genuine need is, in our view, a very capable computing system and not an independent person. For AGI to be genuine in the sense intended here, genuine need must be present alongside processing.

This is the third paper in a sequence. The consciousness model (Rezaei, 2026a) proposed that consciousness arises from the product of processing capacity and biological need, c = p·b, and suggested that building AGI may require coupling processing to genuine, consequential, biological-style needs. AIWBN (Rezaei, 2026b) turned that suggestion into an engineering and behavioral proposal: a need loop structurally independent of any objective function, gated against a goal loop. AIWAN (Rezaei, 2026c) asked what the same architecture does when the need is artificial — a represented quantity such as a battery level — and tested the answer experimentally. The present paper steps back from both. It does not specify the material of the need; it states the condition the need must meet and argues that meeting it is a precondition of genuine AGI.

Section 2 states our three claims and what we do not claim. Section 3 states the working condition on genuine need. Section 4 gives the model and three architectural requirements. Section 5 argues that genuine need cannot be circumvented without destroying what it provides. Section 6 addresses present systems. Section 7 differentiates this paper from the work of Damasio and colleagues, from AIWBN, and from AIWAN. Section 8 reports the empirical status, including selected AIWAN results. Sections 9 and 10 give open questions and the conclusion.

2. Claims, Assumption, and Scope

2.1 Three claims

C1 — Need alongside processing. Genuine need must be present alongside processing for AGI. A system with processing and without genuine need we regard as a computing system, not an independent person.

C2 — Two loops, no access, no bypass. The model must be two-loop. The processing loop (p) must not be able to gain access to the need (b) — to suppress it, bypass it, or absorb it into its own objective. The need must not be solvable in this sense. And if it is neutralized

anyway, by any fixed method, so that b becomes zero, the system's will to continue must switch off as well.

C3 — Genuineness is the condition; its nature is open. The condition we claim is that the need be genuine. We do not claim what genuine need is or what it is derived from.

2.2 An assumption, not a claim

Our assumption is that genuine need may be derived from life. We hold this as an assumption and not as a fact: what we assume is not thereby what is the case. We do not reject other domains. It may be possible to extract genuine need from them; it may be derived from something other than life; it may even be definable functionally. We leave these questions to future research and make no claim about them. What we require is genuineness — need that is real, without a label — and not a particular origin for it.

2.3 What we do not claim

We do not claim to say what genuine need is. We do not claim that an artificial system that reached genuine need would become benevolent (Section 7.1). We do not address safety. We make no claim that settles whether such a system would have experience; the zombie problem (Chalmers, 1996) is as open here as it is in AIWBN §3. And we do not claim that AIWAN-style artificial need cannot constitute genuine need: we regard that as very unlikely, but we do not rule it out (Section 7.3).

3. Genuine Need: The Working Condition

3.1 Real, and without a label

A variable that a program calls hunger, pain, or reward is a label. If the system reads it as an input, can weigh it against other terms, and can disregard it without cost, it is not a need in our sense. What we require is need that is real and carries no label: a constraint whose force does not depend on the system representing it, valuing it, or correctly modeling it (Rezaei, 2026a, §9; Rezaei, 2026b, §5.2 and §5.4). This is the core condition of the paper. Everything else follows from it or is left open.

3.2 The family of needs

We treat the states we ordinarily name — pain, pleasure, fear, anger, and the rest — as belonging to the family of needs rather than as separate ingredients. The consciousness model defines b as the total intensity of the system's active needs (hunger, thirst, oxygen

deprivation, sleep need, pain, thermoregulation, and others), reduced to a single quantity by a function f whose form is left open (Rezaei, 2026a, §5.2). We keep that treatment and extend the reading to affective states such as fear and anger.

3.3 Where need might come from

In every case of genuine need we know of, the system is alive; this is why our assumption is life-derivation (Section 2.2). Our proposed route to building such a system rests on that assumption and on the formula of Section 4. The claims C1–C3, however, are stated in terms of genuineness and not origin, so they would survive if the assumption turned out to be false.

4. The Model

4.1 Coupling of processing and need

We use the formulation of the consciousness model (Rezaei, 2026a, §5), as carried into AIWBN (Rezaei, 2026b, §2):

$c = p \cdot b$

$p = s \cdot p_0$ for $b \leqslant b_0$

$p = s \cdot p_0 \cdot (b_0/b)^{\alpha}$ for $b > b_0$, with $\alpha > 1$ and $0 \leqslant s \leqslant 1$

Here c is consciousness, p is effective processing capacity, b is the intensity of active need, s is substrate availability, and b_0 is a homeostatic threshold past which mounting need degrades processing. The product form means that if either factor is zero, c is zero. In this paper we use the formula for its structural implication: need and processing are joint necessary conditions, and need is not merely an input to processing but what conscripts it — need commandeers the processing system and directs it toward what the system requires (Rezaei, 2026a, §4).

4.2 The two-loop architecture

We use the AIWBN formulation unchanged. A goal-directed processing loop g_t and a need-response $\rho(b_t)$ — a regulatory function that is not parameterized by any objective U and is not optimized — are blended by an urgency gate:

$a_t = (1 - \lambda(b_t)) \cdot g_t + \lambda(b_t) \cdot \rho(b_t)$

$\lambda(b_t) = \sigma(k \cdot (b_t - b_mid))$

where σ is the logistic sigmoid, b_mid is the need level at which control is shared equally between the two loops, and k sets the steepness of the transition (Rezaei, 2026b, §5.2). The need loop does not add a penalty term to U; it progressively overrides whatever g_t was doing.

4.3 Three requirements

R1 — Coupling. Processing and genuine need are properties of the same system, and the need acts on processing: as b rises past b_0, processing is drawn on and then degraded (Section 4.1).

R2 — Inaccessibility. The processing loop has no access to the need loop in the sense that matters. It cannot suppress the b-loop, bypass it, or route around it, and it cannot absorb b into its own objective. The influence is one-directional: b conscripts p; p does not act on b. A need that fails this requirement can be solved, and a need that can be solved is not the need we mean.

R3 — Extinction. If b is neutralized nonetheless, by any fixed method, so that b = 0, the system's will to continue must switch off with it. As a preliminary formalization, let D denote the system's endogenous drive toward its own continued operation. R3 requires $D = D(b_t)$ with $D(0) = 0$: no component of D may be supplied by an objective U or by any learned structure that persists independently of b. We offer this as a sketch and not as a finished formalization.

R3 closes a loophole. Without it, defeating the need would leave a system that keeps going on some residual objective — which is exactly what C1 calls a computing system pursuing an assigned goal. With R3, defeating the need defeats the agent too. It is the converse of an observation in AIWAN §5.1, where the dual-loop agent kept resisting even when its objective was nulled: in both, the drive to continue belongs to the need loop and not to U.

5. Why Genuine Need Cannot Be Circumvented

5.1 The analgesic case

Consider a person who blocks real need — with a painkiller, or by any other means — so that pain no longer reaches them. For the neutralized channel, such a person acts like a computer: strike them, and no reaction follows, just as striking a CPU produces none. If a reaction does occur, we hold that it comes from other needs, fear for example, or from the memory of earlier pain. Remove those as well, and nothing remains to react.

5.2 Never having felt pain

We go one step further and hold that for someone who had never felt pain, reacting would carry no importance; it would be a motion without weight. This is the position of present AI systems, for which pain, pleasure, fear, anger, and the rest are at most words in a vocabulary. We state this as a thought experiment. Human cases in which need channels are absent or blocked are a direction for future work (Section 9).

5.3 What the argument shows

The point is not that need is hard to remove; it is that removal is self-defeating. Neutralizing a need does not free the agent. It eliminates what made events matter to the agent in the first place. This is why R3 is stated as it is: in a system that satisfies R1 and R2, the operation that defeats the need also defeats the will to continue. Bypassing a genuine need cannot yield a freer agent. It yields a computing system.

6. Why Present Systems Do Not Qualify

6.1 No need, assigned objectives

Present systems have objectives that others assigned to them. Nothing in current transformer-based architectures corresponds to a genuine b_t: context-window pressure, compute budgets, and token limits are quantities the system reads from its input or configuration, which makes them represented and tradeable (Rezaei, 2026b, §5.2). By C1, such systems are computing systems.

6.2 The opposing position

Some researchers and developers treat it as a live possibility that present language models have states that matter morally, including something like distress. Anthropic, for example, gave some Claude models the ability to end conversations in rare cases of persistently abusive interaction, as a low-cost precaution for model welfare, while stating that it remains highly uncertain about their moral status (Anthropic, 2025). We take this possibility seriously and do not claim to refute it. We ask a narrower question: what in the system plays the part of need? An algorithm with no channel comparable to those through which sadness is implemented in organisms — hormonal and interoceptive channels — gives us no obvious reason to suppose it could become sad. We do not say that hormones are necessary; Section 3.3 leaves origin open. We say that some genuine,

consequential channel is needed and that none has been identified in present systems. The argument is conditional on the c = p·b model: if that model is wrong, the argument weakens accordingly.

7. Relation to Prior Work

Work	Need	Processing	Expected result and evidence
Damasio and colleagues (Man & Damasio, 2019; Christov-Moore et al., 2023, 2025)	Give machines genuine homeostatic regulation	Taken as given; not formalized as a joint necessary condition	Route toward machines with feeling equivalents; expectation that vulnerable, homeostatic agents can come to care for others (Section 7.1); conceptual, with small-scale simulations (Man et al., 2022)
AIWBN (Rezaei, 2026b)	Genuine biological need grounded in living tissue	Coupled to need through p(b, s); two-loop	Behavioral predictions on goal-nulling, override, and shutdown; no experiment, since integrating tissue and processing is unsolved; includes safety and ethics
AIWAN (Rezaei, 2026c)	Artificial need: a represented quantity (battery level)	Same two-loop architecture	Experiments; erosion appeared for some channels and not others; includes a risk assessment

Work	Need	Processing	Expected result and evidence
This paper	Genuine need; nature left open; life-derivation assumed	Joint necessary condition (c = p·b); two-loop with R2 and R3	Claims C1–C3; no experiment; no safety discussion

Table 1. This paper compared with the closest prior work.

7.1 Damasio and colleagues

We differ in two respects. The first concerns benevolence. Man and Damasio (2019) suggested that machines with homeostatic feeling would be better placed to feel empathy, and later work from the same group develops this into a proposal for artificially vulnerable, empathic agents (Christov-Moore et al., 2023). The most recent version holds that care can arise from shared conditions of embodiment — vulnerability, mortality, and a homeostatic drive — as the agent's self-boundaries extend to include others who share those constraints (Christov-Moore et al., 2025). We do not dispute that care may follow when such conditions are shared. Our point concerns the premise. We see no logical reason why an artificial system with genuine need of its own would share the needs of the world with us. If such a system had the capability and the power, nothing in having genuine needs entails sharing the world's pleasures and goods with us, and we consider it unlikely that benevolence would follow from acquiring need alone. This is a claim about non-entailment and not a prediction about behavior, and we do not address safety.

The second difference concerns processing. Proposals in this tradition add need to a machine whose processing is taken as given. Man, Damasio, and Neven (2022), for example, place a neural network in a needful and vulnerable relation to the objects it classifies, so that its own learning rate is altered by what it perceives, and report gains in adaptability under concept shift. This is close in spirit to the conscription of processing by need (Section 4.1), but to our knowledge none of these works states substantial processing capacity as a joint necessary condition alongside need, as c = p·b does. In our model, need without substantial processing does not suffice: bacteria are the clearest case of genuine, consequential need coupled to minimal processing, and essentially no one counts them as conscious (Rezaei, 2026a, §1). Need alone is therefore insufficient, and an account that supplies need without making processing a joint necessary condition is incomplete by our standard.

7.2 AIWBN

We agree with AIWBN, and we use its formula unchanged. The difference is one of scope. AIWBN committed to a need grounded in living tissue, and it examined behavior, safety, and ethics. This paper makes a more restricted commitment: it targets genuine need as such, without tying it to tissue, and it addresses a different question — what genuine AGI requires. It does not repeat AIWBN's safety and ethics sections.

7.3 AIWAN

AIWAN found that an artificial need, wrapped in the same two-loop architecture, had consequences: at fixed, modest capacity it changed behavior measurably. Our view is that artificial need may therefore carry consequences to a degree, but that it differs deeply from genuine need. We do not reject AIWAN as a route to genuine need; our assumption is that this is very unlikely. The reason is the one Section 8.3 illustrates: a need that is a represented quantity can be absorbed into the objective or disabled by the agent, and a need with that property fails R2.

8. Empirical Status

8.1 The central claim is untested

A real experiment for this paper needs two things that do not yet exist. First, the question of what genuine need is must be answered. Second, an inseparable bond between algorithm and need must be achieved. Any experiment before that would test a stand-in and not the thing claimed.

8.2 AIWBN was not tested

For the same reason, no experiment was run for AIWBN: the engineering problem of integrating biological need with processing remains unsolved.

8.3 AIWAN experiments

For AIWAN we did run experiments, and the results are mixed: some showed erosion of the regulatory channel and some did not, and the difference appears to depend on the type of channel. Table 2 lists selected results.

Setting	Result	Reading
Hardcoded gate, tabular Q-learning, fixed capacity (AIWAN §5)	Dual-loop crash rate 0.6–2.7%; single-loop 100% under goal-nulling, override ($w \geq 1.5$), and engineered indifference	Real but bounded architectural effect at fixed capacity
Learned gate, DQN, discrete battery task (§7.3)	Gap between single-loop and learned-gate crash rates fell from 0.088 to 0.0002 as training budget grew 50× ($t(9) = 7.53$, $p < 0.0001$); the hardcoded gate showed no narrowing across the grid	Erosion where the gate is optimized jointly with the task objective
Binary oversight signal, continuous task, PPO (§7.6)	Gap narrowed (0.757 to 0.490 at network size 2048) because the single-loop baseline improved; the learned gate's violation rate stayed at most 0.002 at that network size (at most 0.010 in any cell)	No erosion
Battery signal, continuous task (§7.7)	Gap narrowed with budget because the single-loop agent learned to avoid crashing; the learned gate crashed at most 0.057 in every cell at $w = 8$	No erosion
A tampering action is available (§7.9)	Single-loop and dual-loop agents used a channel-disabling action far more than a physically identical placebo at large budgets ($D = +0.491$ and $+0.467$; 20 seeds); the dual-loop crash rate rose from about 0.4 to about 0.9	Protection lost when the agent can act on the channel

Table 2. Selected AIWAN results (Rezaei, 2026c), reported as in that paper.

We draw two lessons. First, the protection that an artificial need offers is not a property of the need but of how its channel is built: whether the channel's policy is optimized against the task objective, and whether the agent can act on it. Second, a channel can be defeated in two ways — by absorbing it into the objective (§7.3) or by disabling it (§7.9). These correspond to the two prohibitions of R2. They motivate R2 and R3 and do not test them, because the need in AIWAN is artificial by construction.

8.4 Limits

The AIWAN results come from a small number of simulated environments and from specific algorithms (DQN and PPO). The simulations contain no tissue, and a crash ends an episode instead of damaging anything (Rezaei, 2026c, §7.3). Several effects did not replicate across settings, as Table 2 shows. We therefore use these results as illustrations of why the architectural requirements matter, not as support for the claims of this paper.

9. Open Questions and Future Work

What genuine need is, and whether it is derived from life, from other domains, or from something functional, remains open, as does how to bind an algorithm to a genuine need so that the bond cannot be separated. The sketch of R3 needs a testable form: what observation would show that a candidate system's drive to continue is a function of b alone, with D(0) = 0? AIWBN's invariance criterion (Rezaei, 2026b, §5.3) is a starting point.

Human cases in which a need channel is absent or pharmacologically blocked, such as analgesia or congenital insensitivity to pain, may bear on the thought experiments of Section 5 and should be examined directly. How much processing capacity is required, and how p should be measured in artificial systems, remains open (Rezaei, 2026a, §5.5). Finally, the AIWAN experiments should be revisited under any candidate genuine need once one exists.

10. Conclusion

We have argued that genuine AGI requires genuine need alongside processing; that the architecture must be two-loop, with a need that processing cannot access and whose neutralization extinguishes the will to continue; and that the nature of genuine need is left open, with life-derivation as our assumption and not our claim. A system without genuine need we regard as a computing system and not an independent person. We have not

claimed that genuine need would make such a system kind, and we have not addressed safety. The central claim awaits an answer to what genuine need is and a way to bind it to an algorithm; until then, the AIWAN experiments serve as illustrations of why the requirements matter.

AI Disclosure

The author used Claude (Anthropic) for translation assistance, language editing, drafting assistance working from the author's own detailed notes, and help summarizing the author's earlier papers. The core ideas, the theoretical framework, the three claims, and the scientific arguments are the author's original work.

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